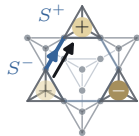


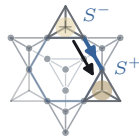
i.a) create


 $S_i^+ S_j^-$: pair splits

i.b) move


 $S_i^+ S_j^-$: spinon walks

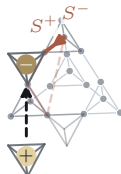
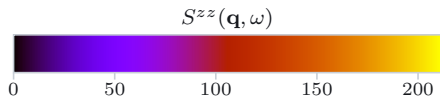
i.c) annihilate


 $+$ meets $-$
 $\mathbf{q}_\gamma = (0, 0, 0)$

i.d) create


 $S_i^+ S_j^-$: pair splits
i.e) apply $S_i^+ S_j^-$ 
 $+$ carried out of the cell

i.f) identify


PBC maps it onto $-$
 $\mathbf{q}_\gamma = (0, 0, -1)$


iii.a) periodic ED



iii.b) winding-free

